

DILLINGHAM, AK, USA

WMO: 703210

Lat: 59.050N

Lon: 158.517W

Elev: 86

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25513

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-12.9	-26.9	1.2	-15.1	-23.7	1.5	-10.7	31.7	19.7	27.7	21.5	3.4	350	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	70.6	58.8	66.8	57.4	64.0	55.5	60.6	67.9	58.7	65.0	57.0	62.4	9.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	70.2	62.4	55.4	65.5	60.3	54.3	62.9	59.2	26.8	68.4	25.4	65.0	24.4	62.2	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.0	21.7	19.1	DB	-22.4	76.1	7.3	4.8	-27.7	79.5	-32.0	82.3	-36.1	85.0	-41.4	88.4	(4)
(5)			WB	-21.7	63.3	6.4	3.8	-26.3	66.0	-30.1	68.2	-33.7	70.3	-38.3	73.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.3	16.0	20.7	21.4	33.0	43.7	51.8	55.4	54.4	48.0	36.3	23.8	18.2		
	DBStd	17.34	15.12	13.22	11.32	7.92	6.21	4.91	4.63	4.43	4.89	8.15	11.48	14.29		
	HDD50	5815	1055	819	885	510	213	32	5	10	89	425	786	986		
	HDD65	10843	1520	1239	1350	960	661	398	301	328	510	889	1236	1451		
	CDD50	452	0	0	0	0	17	85	172	148	30	0	0	0		
	CDD65	4	0	0	0	0	0	1	3	0	0	0	0	0		
	CDH74	42	0	0	0	0	2	13	23	4	0	0	0	0		
	CDH80	5	0	0	0	0	0	2	3	0	0	0	0	0		
(14) Wind	WSAvg	8.5	10.2	10.1	9.2	8.8	8.2	7.1	7.0	7.0	7.6	8.6	8.9	9.5		
(15) Precipitation	PrecAvg	33.2	2.8	2.3	2.1	1.7	1.9	2.2	2.9	3.2	4.0	3.4	3.4	3.2		
	PrecMax	48.0	6.5	4.6	6.1	3.1	4.2	3.7	4.8	5.2	8.0	6.2	7.7	7.2		
	PrecMin	21.3	0.6	0.4	0.2	0.3	0.3	1.1	1.5	1.3	0.8	0.3	0.5	0.7		
	PrecStd	6.1	1.3	1.1	1.4	0.7	0.9	0.6	0.8	1.1	1.5	1.3	1.7	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.1	39.2	42.9	54.4	69.9	74.8	76.9	72.3	63.3	53.8	45.8	40.8		
		MCWB	36.4	36.3	37.7	43.6	52.4	58.9	63.3	60.8	53.4	48.3	42.2	37.2		
	2%	DB	37.2	37.2	37.7	49.9	63.1	69.5	71.6	67.9	59.9	50.9	43.5	38.8		
		MCWB	34.7	34.9	33.8	41.7	49.7	56.4	60.5	59.2	52.9	47.1	40.7	35.9		
	5%	DB	36.3	36.4	36.5	45.9	57.8	64.6	67.3	64.4	57.1	48.4	40.8	36.9		
		MCWB	34.1	34.2	33.3	39.6	47.3	55.2	58.8	56.6	51.2	45.5	38.4	34.5		
	10%	DB	34.7	35.5	35.5	43.2	54.4	61.9	64.0	62.6	55.0	46.4	37.6	35.8		
		MCWB	32.9	33.5	33.0	37.9	45.9	53.2	56.4	55.5	50.4	43.6	35.4	34.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.3	37.6	38.4	45.4	54.3	60.9	66.0	62.7	57.3	50.1	43.9	38.2		
		MCDB	38.4	38.2	41.8	52.4	65.8	70.2	73.6	69.5	60.5	51.8	44.8	40.3		
	2%	WB	35.7	36.0	35.3	42.0	51.0	58.1	61.6	60.1	54.7	48.0	41.3	36.6		
		MCDB	37.0	37.0	37.1	48.1	60.9	66.6	69.2	65.9	57.3	49.9	44.0	38.3		
	5%	WB	34.1	34.3	33.9	40.3	48.6	56.1	59.2	58.2	53.0	46.2	39.0	35.0		
		MCDB	35.7	35.7	35.8	45.5	56.5	63.6	66.4	63.0	55.6	47.9	40.8	36.6		
	10%	WB	32.8	33.0	32.7	38.5	46.5	54.0	57.3	56.6	51.3	44.5	36.4	33.7		
		MCDB	34.5	34.7	34.7	42.6	52.7	60.5	63.0	60.9	54.1	46.3	37.9	35.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	10.9	12.5	12.0	13.9	13.8	12.1	11.7	11.7	10.5	10.9	11.0		
		MCDBR	7.2	8.0	9.4	15.5	21.5	20.1	19.5	16.3	14.3	9.9	8.2	7.2		
	5% WB	MCWBR	6.3	7.1	7.4	9.9	12.5	11.3	10.9	9.7	9.0	7.4	7.4	6.8		
		MCWBR	7.1	7.9	8.4	14.4	19.0	18.2	17.4	14.0	11.3	8.3	8.1	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.299	0.312	0.355	0.393	0.361	0.364	0.369	0.332	0.307	0.289	0.259		
	taud		2.121	2.180	2.186	2.189	2.228	2.421	2.452	2.431	2.512	2.425	2.095	1.987		
	Ebn at Noon		172	225	260	267	265	277	273	263	257	226	151	118		
	Edh at Noon		15	24	32	38	40	33	32	30	24	19	15	12		
(44) All-Sky Solar Radiation	RadAvg		129	330	895	1396	1638	1668	1421	1147	802	416	167	74		
	RadStd		24	64	141	171	232	146	150	141	104	51	25	9		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page